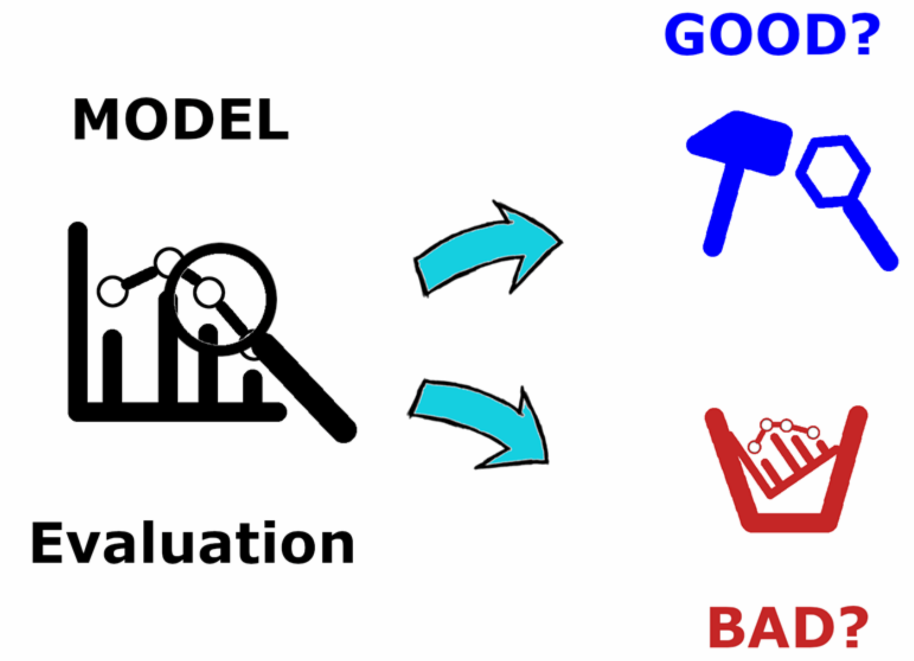
Model Evaluation for Real-World IoT

Why Evaluation Matters

✓ Ensures reliability in real environments
✓ Validates performance beyond the lab

From Lab Success → Field Reality

- A model that performs well in testing may fail in real-world environments.
- **Reasons**
  - ✓ network delays
  - ✓ device limitations
  - ✓ noisy sensor data
  - ✓ environmental variability



The best AIoT model is not the most accurate — it is the most reliable and practical

Accuracy Alone is NOT Enough

Traditional ML Focus

✓ Accuracy
✓ Precision & Recall

AIoT Requires More

✓ real-time performance
✓ reliability
✓ resource efficiency

Key Evaluation Metrics for AIoT

Performance Metrics

- Accuracy & F1 Score
- Precision & Recall

Operational Metrics

- Latency
- CPU & memory usage
- Power consumption

Performance Evaluation in real environments

Latency: Real-Time Response Matters

- ⚡ Autonomous systems require instant decisions
- ⚡ Safety systems need millisecond response
- ⚡ Industrial control must react immediately

Resource Usage & Edge Constraints

Edge Devices Have Limits

⚠ Limited CPU power
⚠ Limited memory
⚠ Battery constraints
⚠ Intermittent connectivity

Evaluation Must Include

✓ model size
✓ inference speed
✓ energy efficiency

Performance Evaluation in real environments

False Alarms vs Missed Events

- **Operational Trade-Off**
  - Too many false alarms → alert fatigue
  - Missed detections → safety & financial risk
- **Important Metrics**
  ✓ Precision
  ✓ Recall
  ✓ False Positive Rate

Robustness & Reliability

- **Models Must Handle**
  ✓ sensor noise
  ✓ missing data
  ✓ environmental changes
  ✓ device failures
- **Robust Systems Continue Operating** even with imperfect inputs.

Lab Testing vs Real-World Evaluation

- **Lab Testing**
  - ✓ clean datasets
    ✓ controlled conditions
- **Field Evaluation**
  - ✓ real sensor noise
    ✓ temperature & vibration effects
    ✓ network disruptions

Edge vs Cloud Model Tradeoff

- **Complex Cloud Model**
  ✓ high accuracy
  ❌ higher latency
  ❌ connectivity dependency
- **Lightweight Edge Model**
  ✓ real-time decisions
  ✓ reliable operation
  ✓ offline capability

Transition: Choosing the Right Model

Key Takeaway

- The best model is not the most accurate one.
- It is the one that performs reliably in real conditions.